Announcements

Midterm 1 grades released + sol'ns posted

Mean: 65.2/86
Median: 64/86
Std. dev.:

Q1: 92%
Q2: 82%
Q3: 81%
Q4: 87%
Q5: 46%

Gradelines:

A/A-: 70-85
B+/B/B-: 55-69.9
C+/C/C-: 38-54.9
D+/D/D-: 15-37.9

out of 86

Gradeline calculator posted

Regrade requests open for 1 week

No quiz today

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§6.5: Generalized Permutation and Combinations

$P(n,k)$ and $\binom{n}{k}$ refer to permutations/combinations <u>without</u> repetition and with <u>distinguishable</u> objects

e.g. ABCDEF

BCAE
4-permutation

$\{C, E\}$
2-combination

If we allow repetition, we allow examples like

BBBB	$\{c, c\}$
4-perm w/ rep.	2-comb. w/ rep.

For a set of size n, the number of

r-perms w/ repetition is	r-combs w/ repetition is
n^r	$\binom{n+r-1}{r}$ (*)
(by prod. rule)	

Idea behind (*): "sticks and stones".

Stones are the elements

Sticks are the "separators"

Ex 4: 4 different kinds of cookie. How many different ways are there to choose 6 cookies <u>with</u> (potential) repetition?

e.g.

$$**|*||***$$

2 of type 1

1 of type 2

0 of type 3

3 of type 4

6 stars (cookies)

4 types, so $4-1=3$ separators

Choose the spots for the 6 cookies (or 3 separators)

Num ways:

$$\binom{6+3}{6} = \binom{9}{6} = \frac{9\cdot 8\cdot 7}{3\cdot 2\cdot 1} = 84$$

Ex 5: How many nonnegative integer sol'ns does the eqn. $x_1 + x_2 + x_3 = 11$ have

11 "stones" $3-1=2$ "sticks"

$$** | *** | ******$$

$$2 + 3 + 6 = 11$$

Num ways: $\binom{11+(3-1)}{11} = \binom{13}{11} = 78$

Permutations of partially indistinguishable objects:

e.g. AABBBC BABCBA

For n total objects, k types, n_i of the ith type, there are $\dfrac{n!}{n_1!\,n_2!\cdots n_k!} =: \dbinom{n}{n_1,\dots,n_k}$ permutations (**)

(Note: $\dbinom{n}{k,n-k} = \dbinom{n}{k}$)

Idea: take a permutation:

$n!$ ways

BABCBA

Swapping around the first type of object doesn't change the permutation

$n_1!$ ways to do this

Divide, by division rule

Same for $n_2, n_3, \ldots$

all the same

BABCBA
BABCBA
BABCBA
BABCBA
BABCBA
BABCBA

Next, we want to put n objects into k boxes

Distinguishable objects into distinguishable boxes:

k^n ways

If we want $n_1, n_2, \ldots, n_k$ elts. in box $1, 2, \ldots, k$:

$\binom{n}{n_1, \ldots, n_k}$ ways

Indistinguishable objects into distinguishable boxes:

Sticks and stones: $\binom{n+k-1}{n}$

Other two cases are harder: use ad hoc methods

Distinguishable objects and indistinguishable boxes:

Ex 10: How many ways are there to put four (distinguishable!) students into at most three groups?

Sol'n:

All four in one group: 1 way ABCD

3 in one group, 1 in another: 4 ways ABC, D ACD, B ABD, C BCD, A

2 in one group, 2 in another: 3 ways AB, CD AC, BD AD, BC

2 in one group, 1 in another, 1 in a third: 6 ways

AB, C, D BC, A, D
AC, B, D BD, A, C
AD, B, C CD, A, B

not 6 since BC, AD is the same as AD, BC

Total: $1+4+3+6=14$

Indistinguishable objects into indistinguishable boxes:

Ex 11: How many ways are there to pack 6 identical copies of a book into (at most) 4 identical boxes?

Sol'n: List the possibilities:

6,0,0,0 (the same as 0,6,0,0)

5,1

4,2

4,1,1

3,3

3,2,1

3,1,1,1

2,2,2

2,2,1,1

boxes listed in descending order

"integer partitions"